

Lightweight Model Design for Efficient Bone Suppression Image Transformation

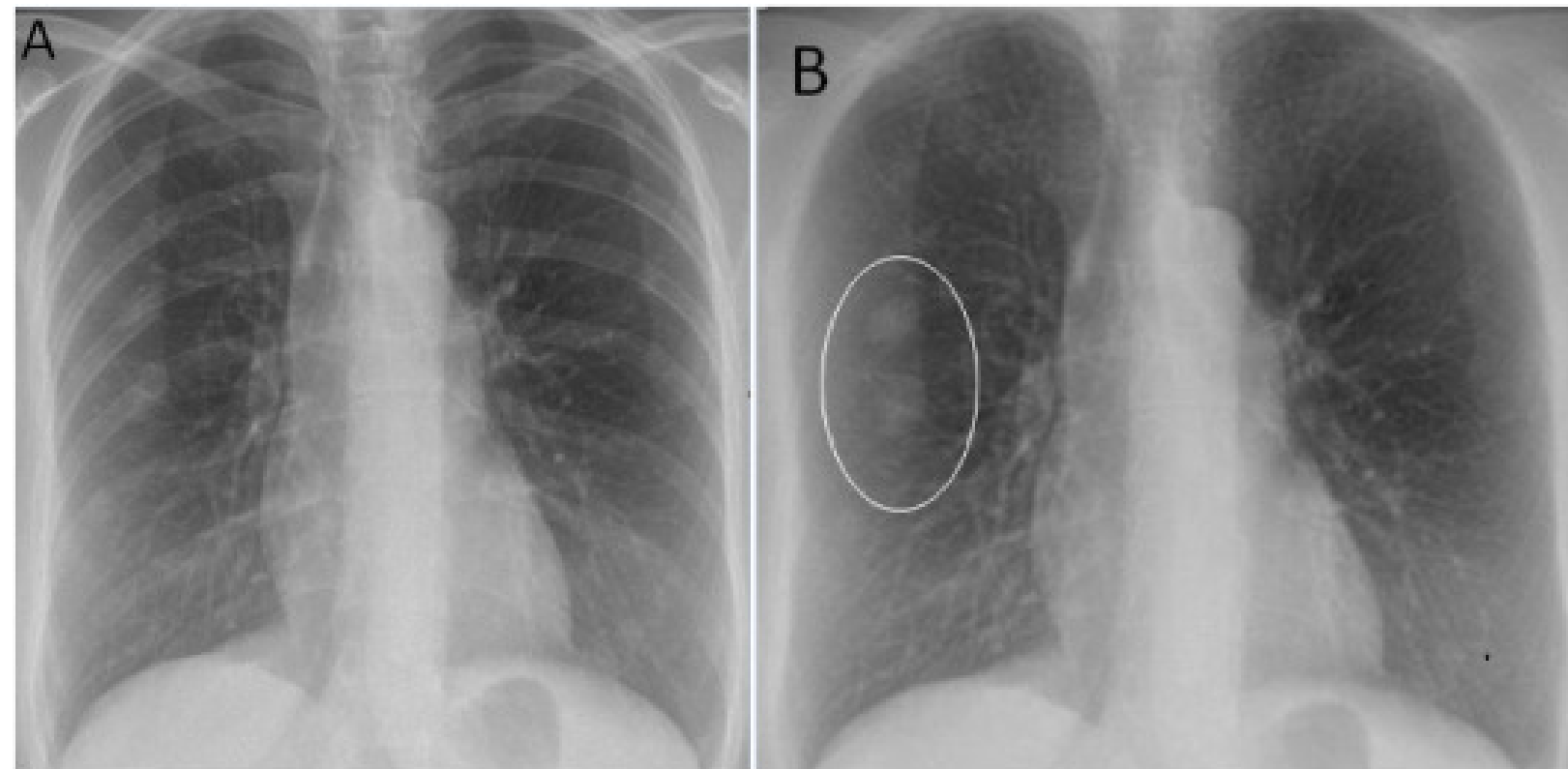
Lee sangjun

Introduction

This topic covers one of the important tasks in medical imaging analysis: Bone suppression, a technique for suppressing bone information in medical photographs.

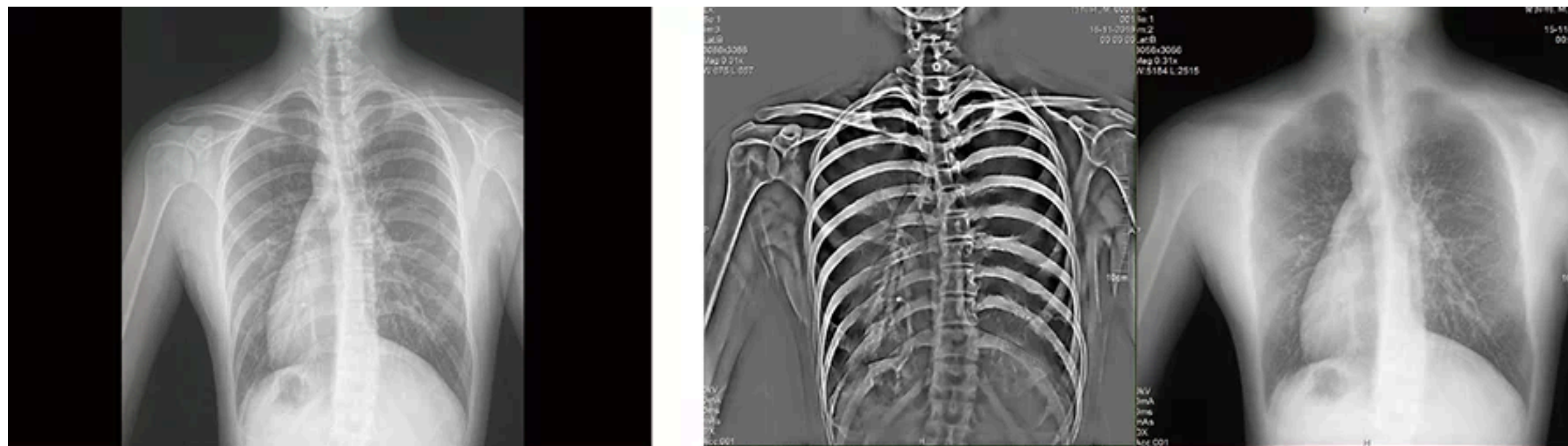
This is especially essential for improving the accuracy of detection of microscopic lesions, e.g., lung nodules or tumors, in chest X-rays.

Effective Bone suppression plays a very important role in the clinical field because detection of lesions becomes difficult when bone structures cover lung tissue.



Traditional Bone Suppression methods have relied heavily on traditional image processing techniques such as filtering and image separation, or physical image acquisition methods (e.g., dual-energy subtraction).

However, these methods are machine expensive or difficult to apply clinically, and require a lot of additional processing to achieve high-quality results.



Original Image

Bone Tissue

Soft Tissue

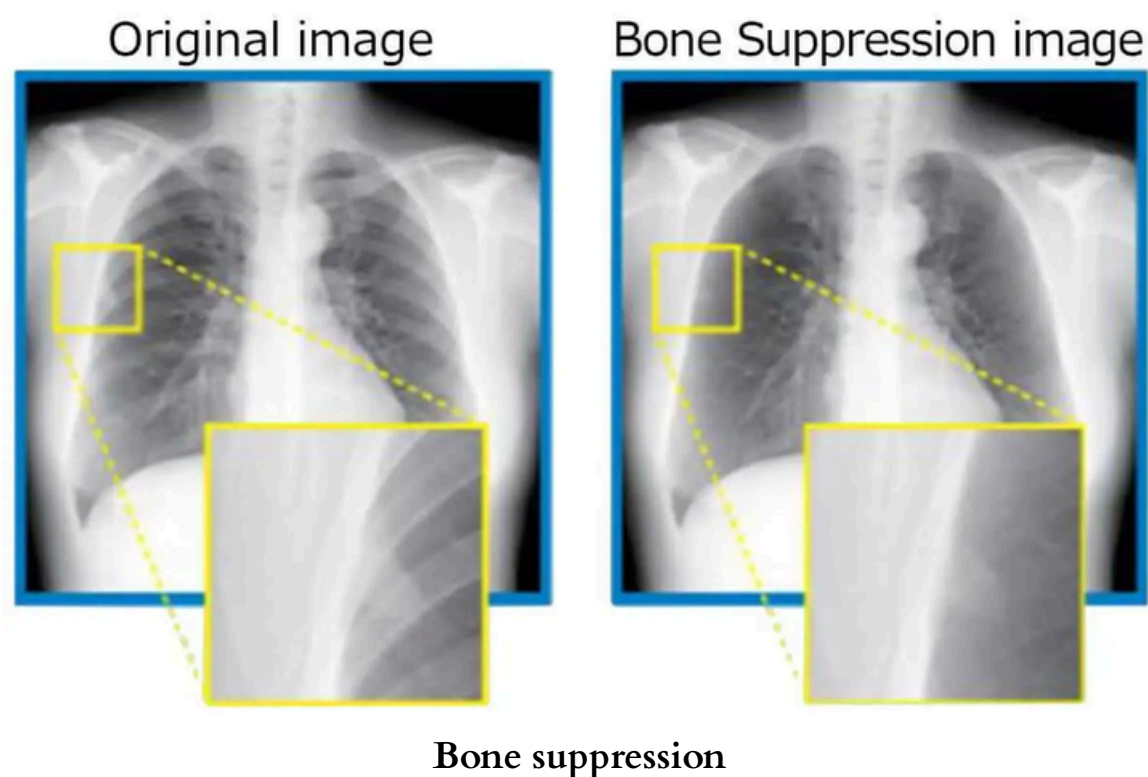


Applications of Image-to-Image Translation

Medical images are used for diagnostic assistance (Bone suppression, lesion diagnosis, etc.) and data augmentation through modality transformation: changes and transformation between different medical images such as MRI, CT, and X-ray.

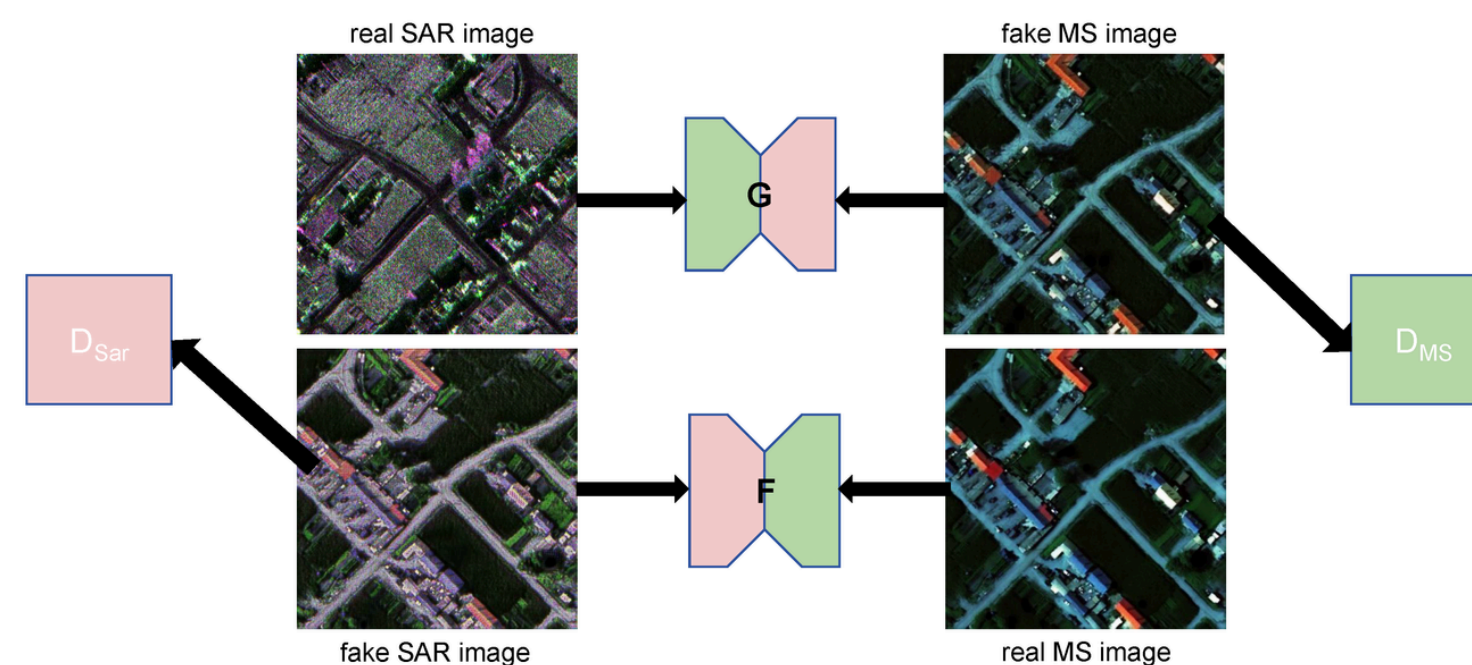
In addition, it can be used in a variety of places, including satellite and remote sensing, style transformation, robotic and autonomous pedestrian detection enhancement technologies, augmented reality (AR) and virtual reality (VR).

Healthcare & Medical Imaging



<https://www.konicaminolta.com/global-en/healthcare/products/healthcare-it/bs/index.html>
<https://www.mdpi.com/2072-4292/16/21/4045>
<https://arxiv.org/pdf/2209.03625>

Satellite & Remote Sensing



Map making and updating: convert satellite images into map images or use them for change detection.

Robotics & Autonomous Driving

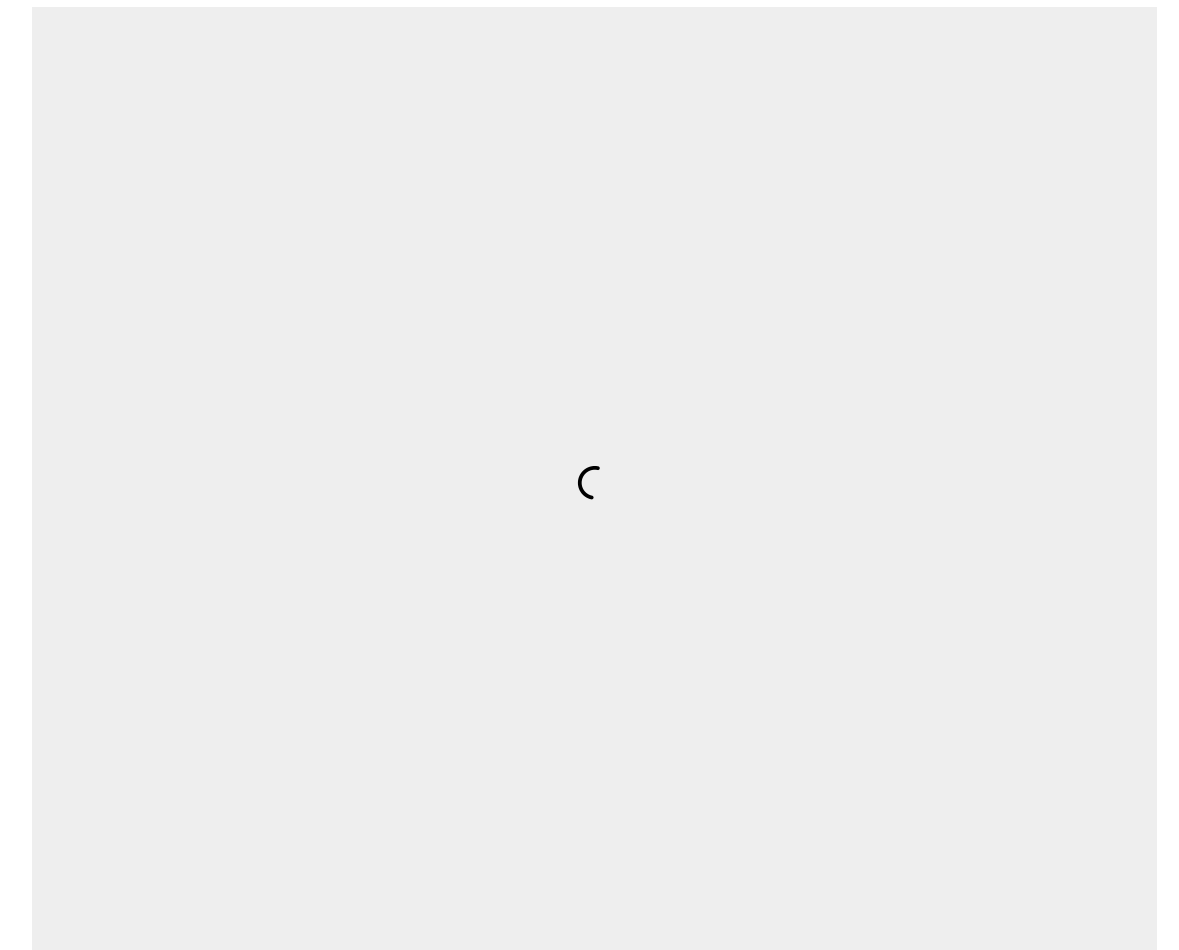
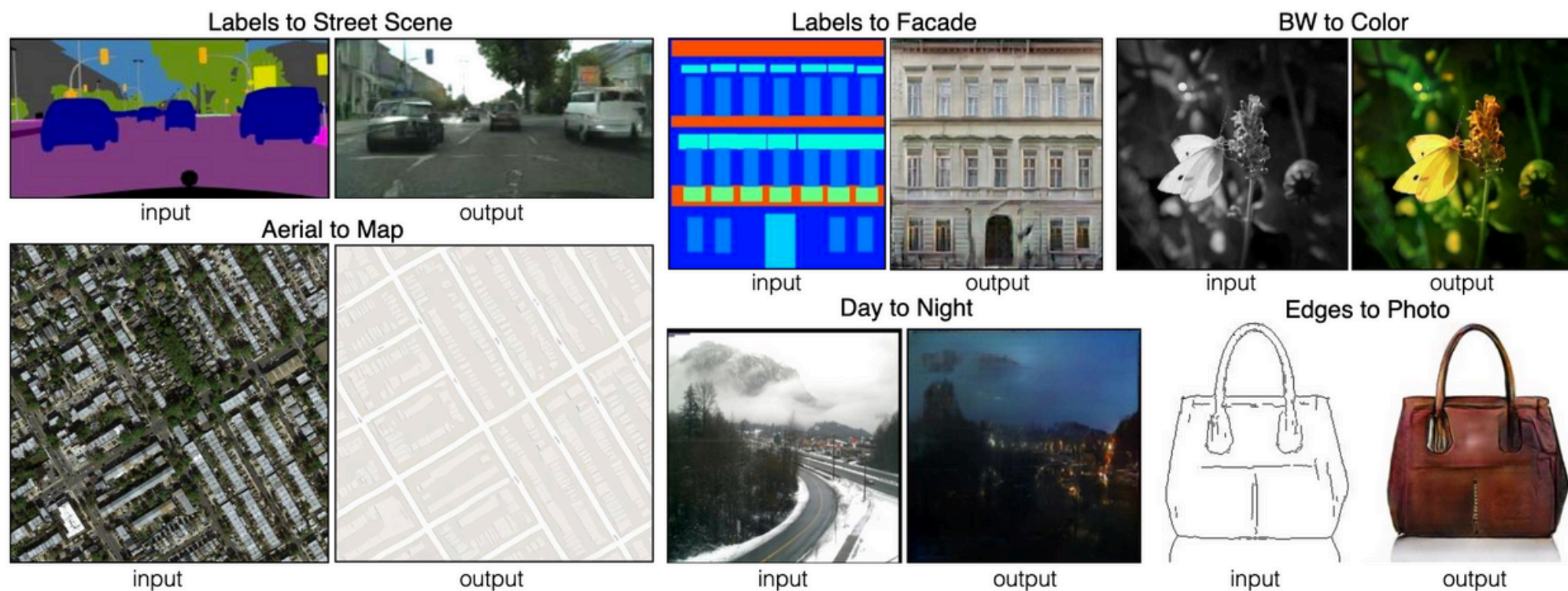


Pedestrian detection

Pix2Pix

Pix2Pix is a model published in the paper "Image-to-Image Translation with Conditional Adversarial Networks" and is an Image-to-Image Translation model based on a conditional GAN model.

cGAN is a generative model of a simply extended version of the conditional model such that discriminator D and generator G are conditioned by additional information y , such as class labels.



The generator acts like a painter who creates the desired output image based on the input image. The discriminator acts as an appraiser to determine whether the generated image is real or fake.

The two learn by competing with each other, where the generator tries to produce more sophisticated images to deceive the Discriminator, who develops sharper discrimination to see through the tricks of the Generator.

Discriminator adopts a unique structure called **PatchGAN**.

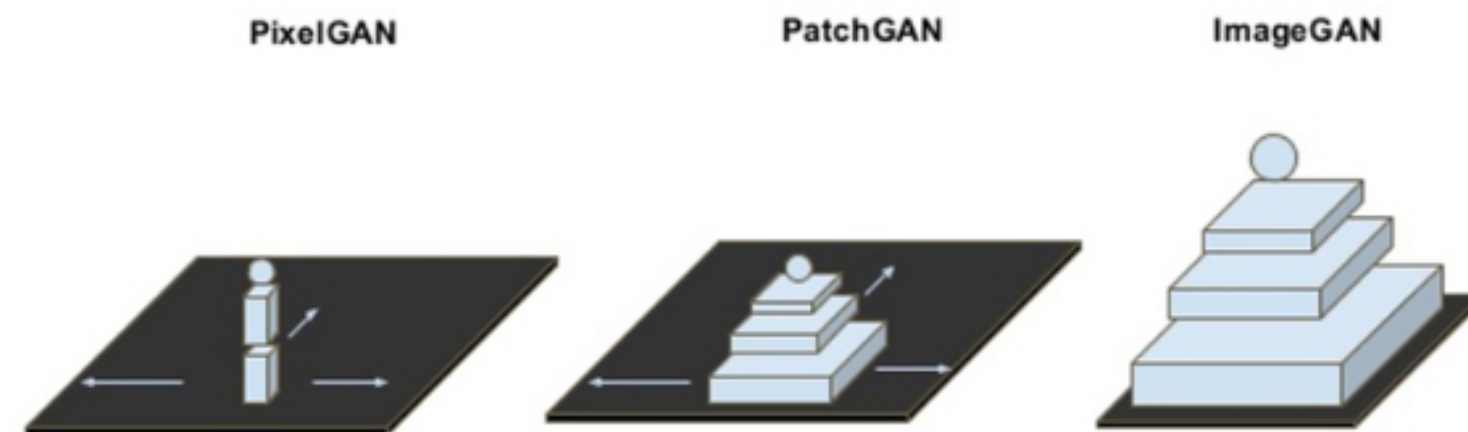
Instead of determining the entire image at once, divide the image into several smaller patches to determine whether each patch is real or fake.

This helps to provide a more detailed analysis of the local features of the image.

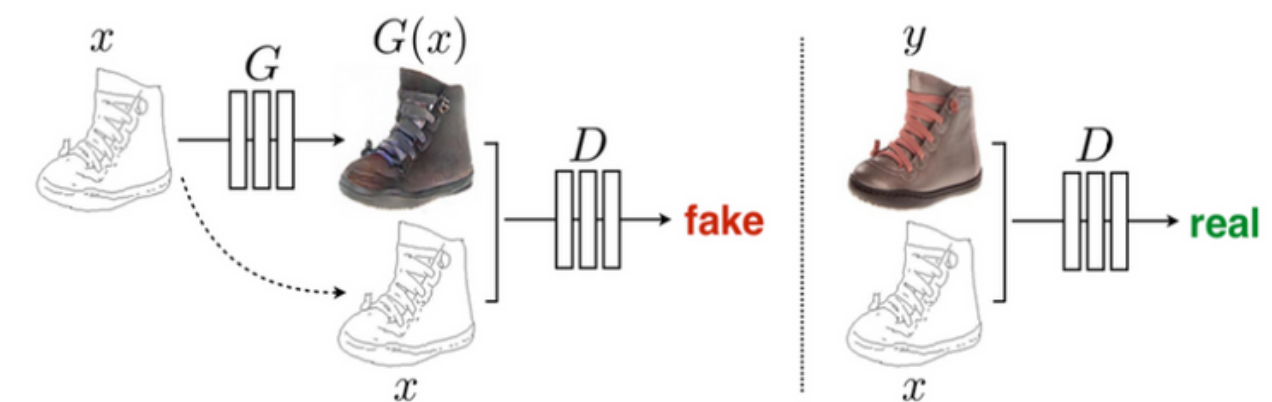
- For example, input data: When you want to convert a graphic image into an image in the form of photo, unlike the GAN model without conditions, both generator and discriminator observe input graphic x .
- The **generator** tries to produce images under given conditions using **U-Net**, an encoder-decoder with skip connections between mirrored layers.
- The **discriminator** uses a convolution-based **PatchGAN** classifier to try to classify whether each patch is real or fake in an image patch $N \times N$ scale structure.

However, existing image conversion models require a lot of memory and computation for high performance. There are many difficulties in applying the model, especially when dealing with large images or when the server environment is limited.

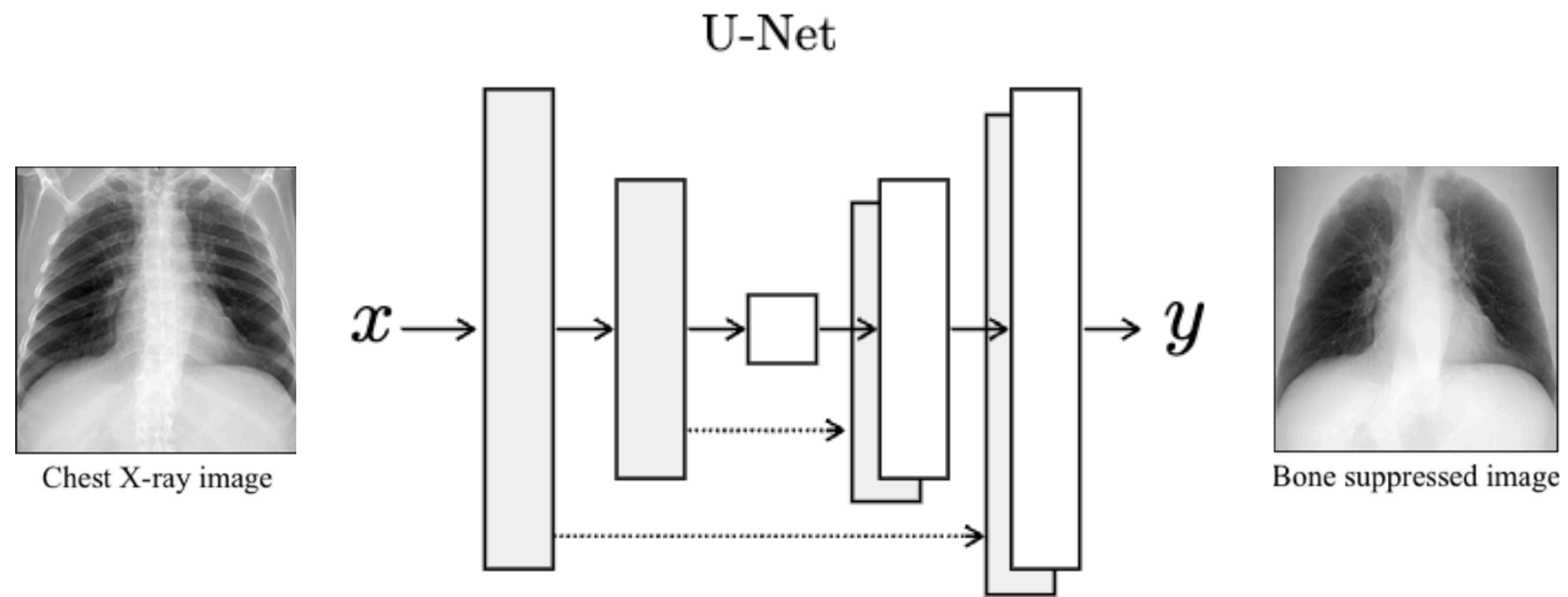
To address these challenges, a new lightweight model is required.



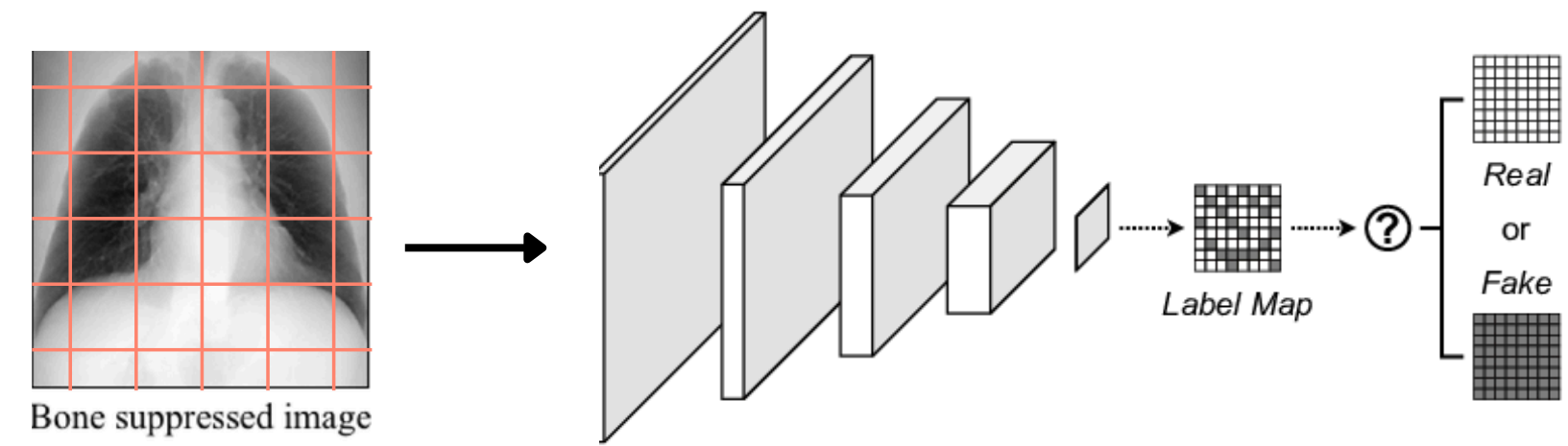
Discriminator uses a convolutional PatchGAN classifier, which only penalizes structure at the scale of image patches $N \times N$, so it tries to classify if each patch is real or fake.



Generator : Lightweight U-Net



Discriminator : PatchGAN



Experiments : Train data

Original 007.png



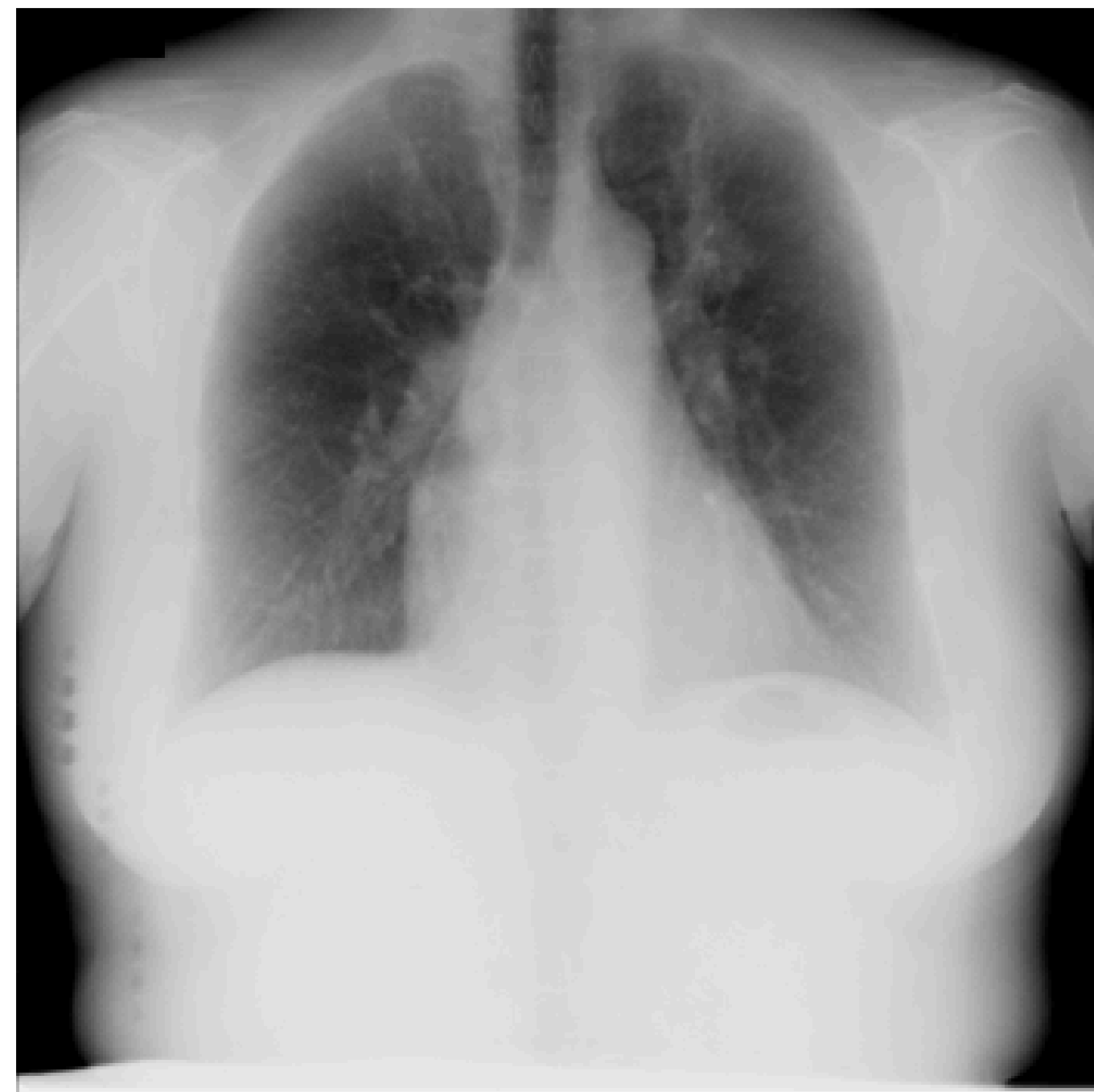
Output7.png



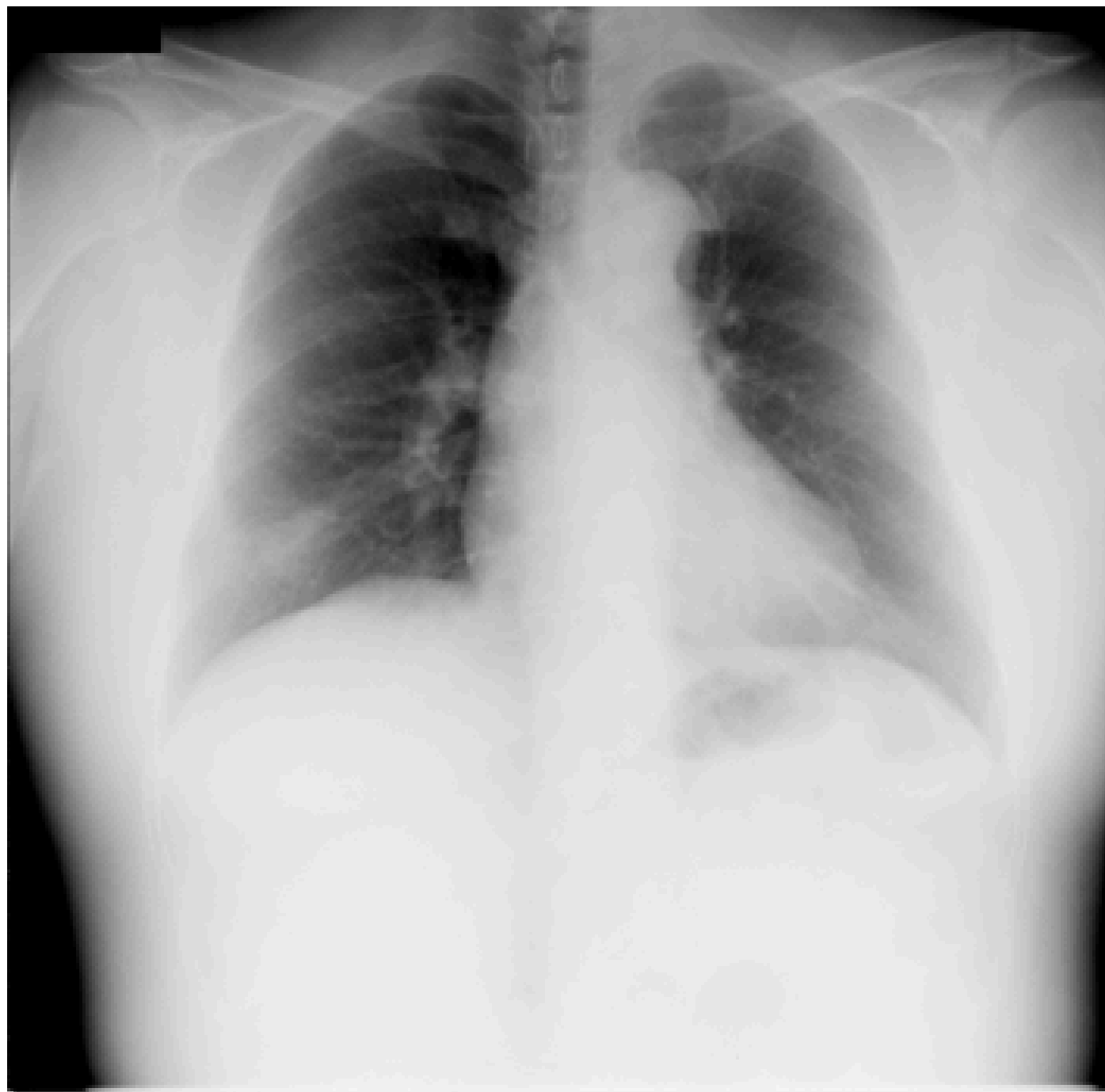
Original BS 003.png



Output3.png



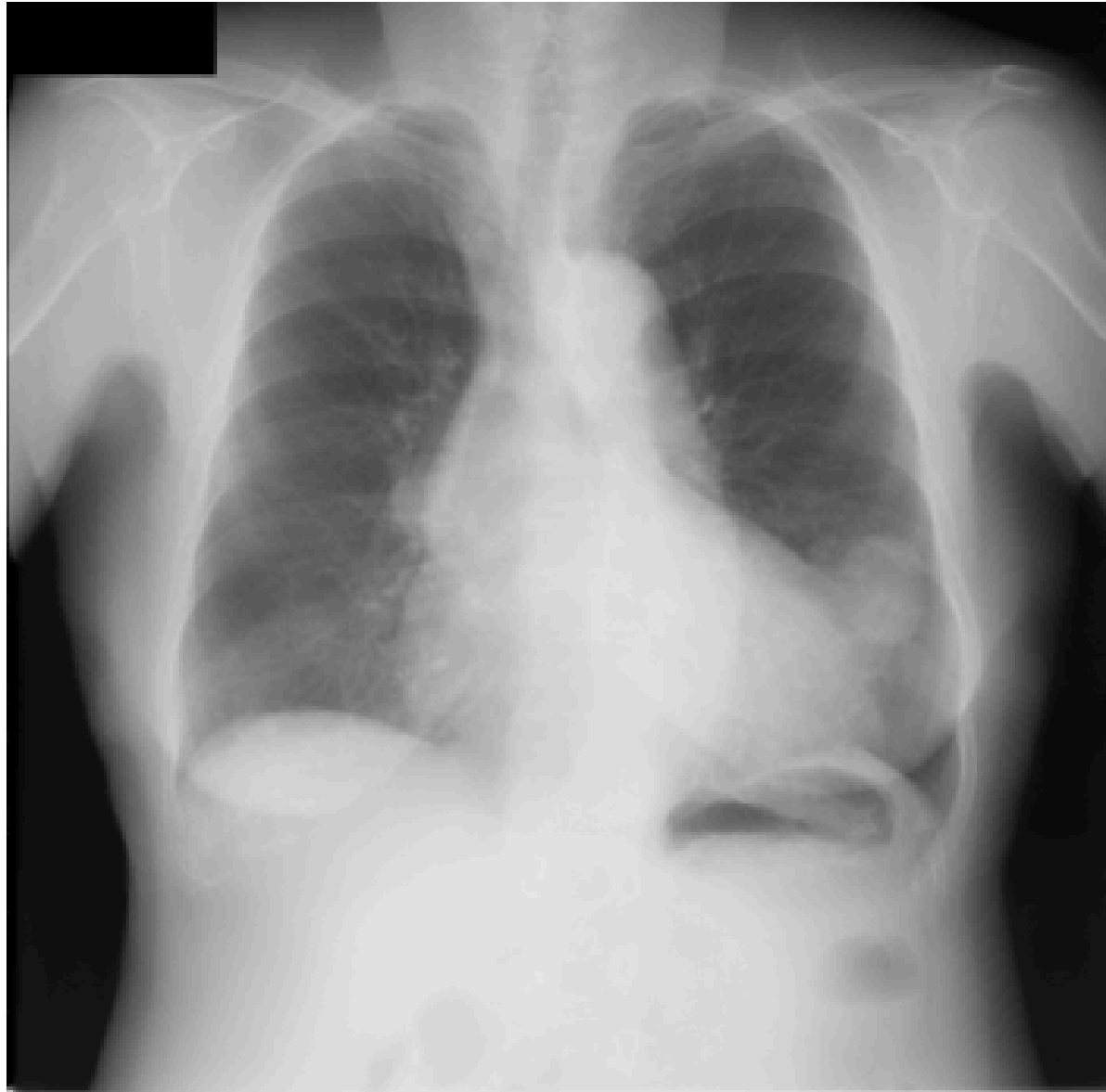
Original 41.png



Output_41.png



Original 002.png

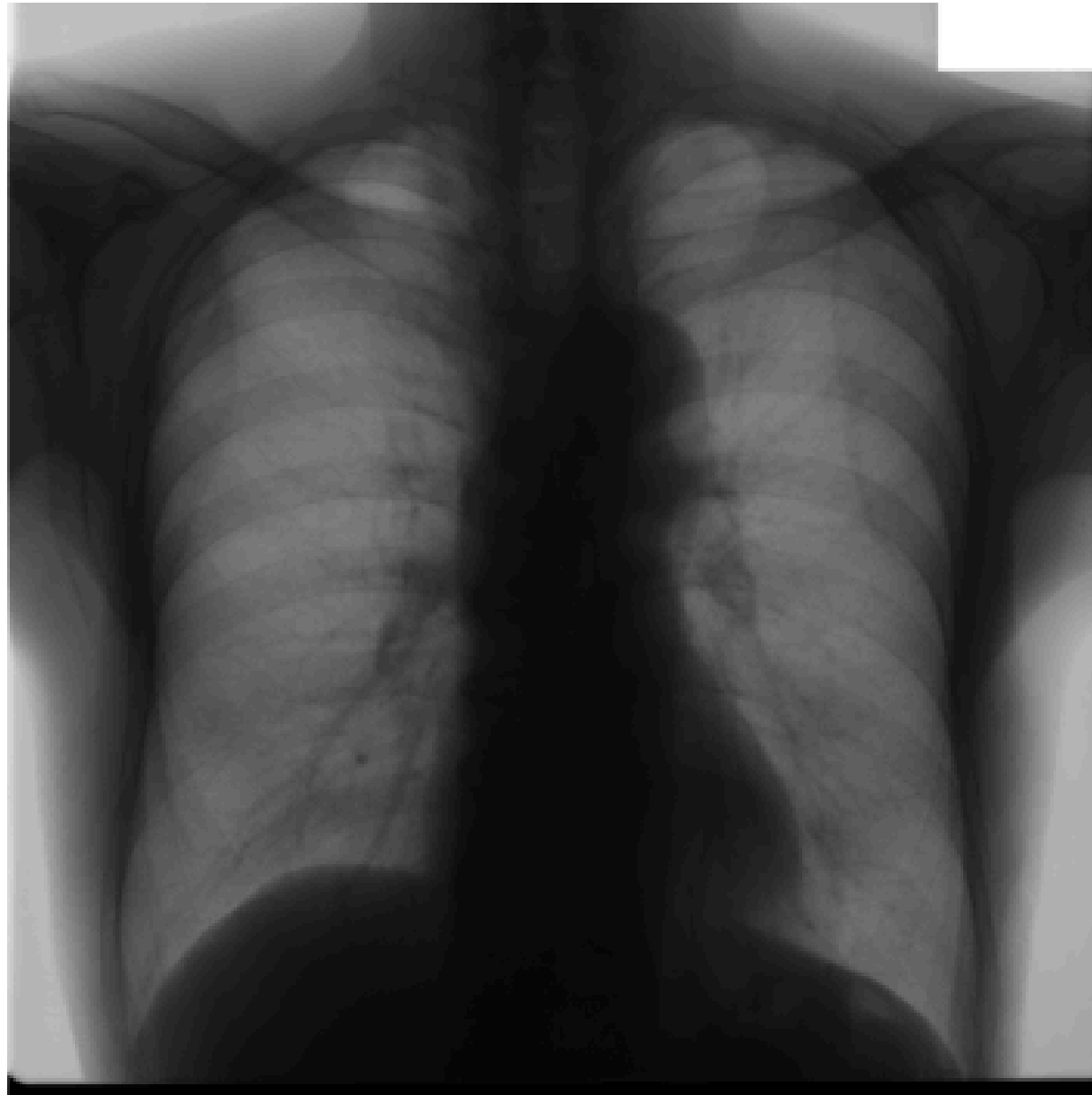


Output2.png

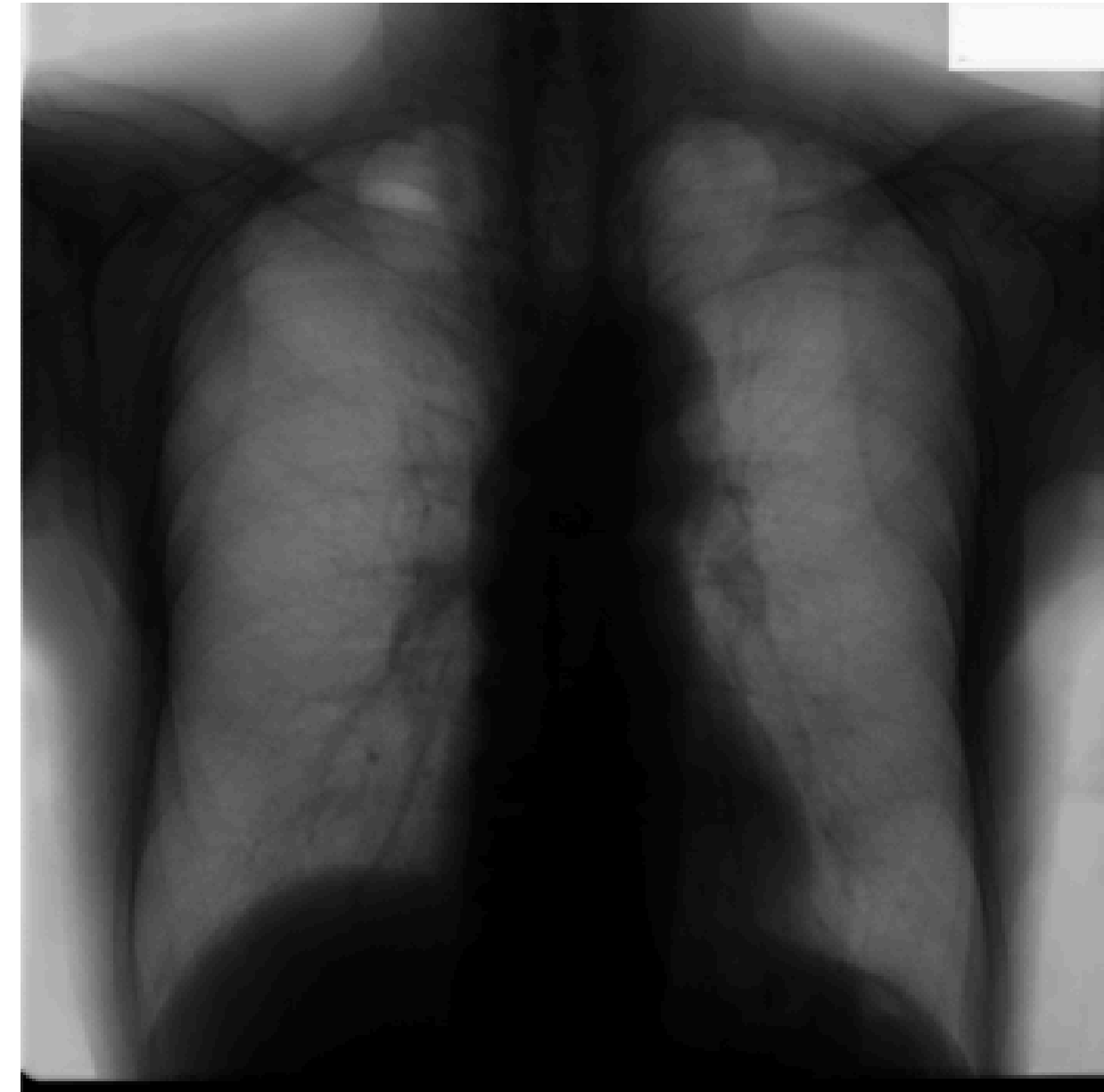


Experiments : Data generalization

Original 002.png



Output2.png



Original d 14.png



Output_14.png



Original 161510.png



Output_161510.png



Trained	Model 1	Model 2	Model 3	Model 4
SSIM	0.9807	0.9835	0.979	0.9763
PSNR	40.6908	40.9663	39.985	38.0264

Discussion : Improve Generalization Performance

Validation on more diverse and large datasets is required for practical clinical applications.

In the future, we can study various methods to improve generalization performance (the possibility of introducing Unpaired Learning using **CycleGAN** or **Transformer-based models**), as well as how to link **segmentation** and **disease detection**."

Conclusion

Bone suppression is a key technology to improve the reading accuracy of chest images. The application of the latest Image-to-Image Translation models, such as Pix2Pix, is taking medical AI's potential for real-world clinical use to the next level.

QnA